

RISK & RETURN IN INVESTING

Financial Mathematics • CA-III • Unit V and Unit VI

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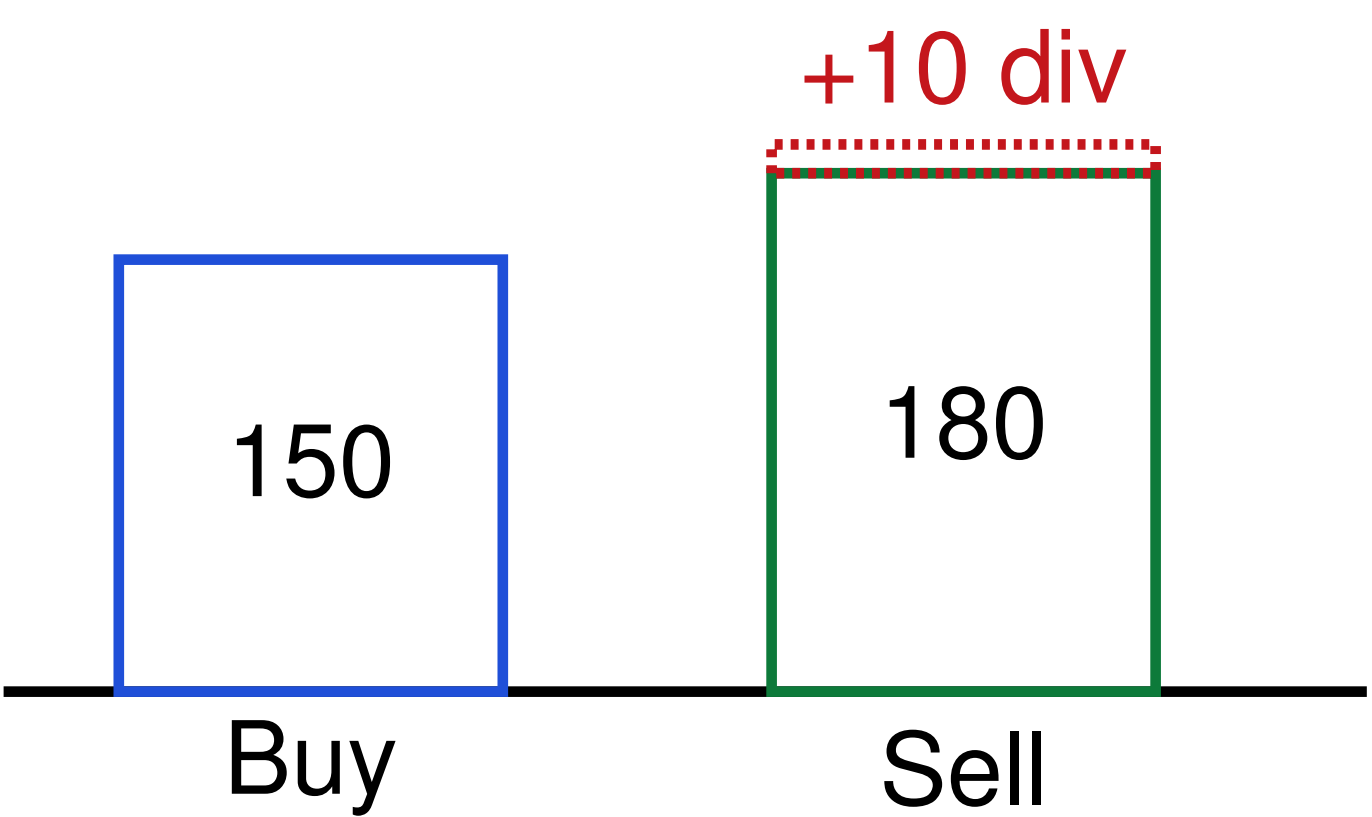
Division: **AIML-A**

UNIT V: BASICS OF INVESTMENT

1. Investment Return

$$\text{HPR} = \frac{\text{Income} + V_f - V_i}{V_i} \times 100$$

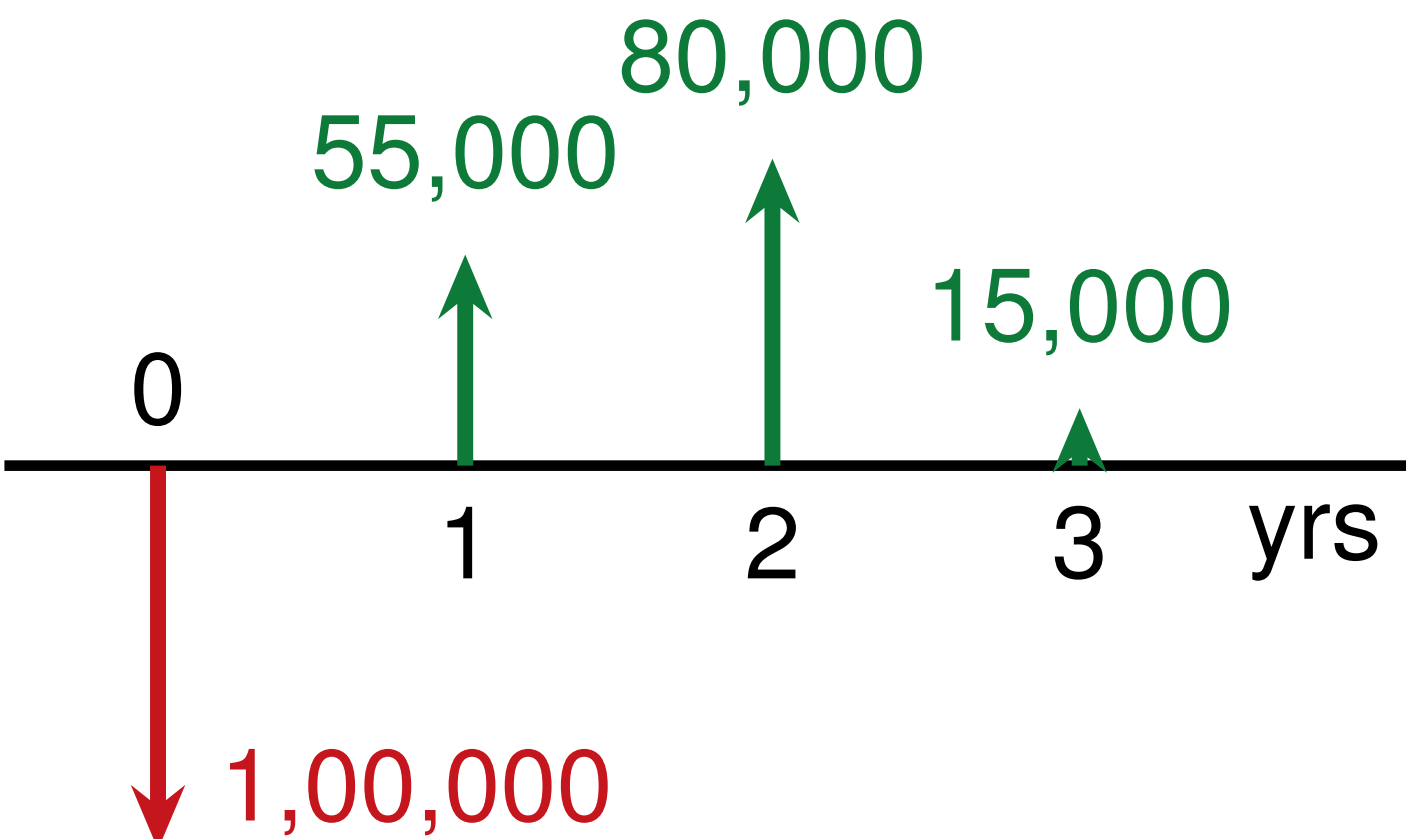
Buy Rs 150, sell Rs 180,
dividend Rs 10:
HPR = $40 \div 150 = 26.67\%$



2. Uneven Cash Flows

$$\text{NPV} = \sum C / (1 + k)^t - I$$

$I = \text{Rs } 1,00,000, k = 10\%$
(PVIF to 3 decimals)



NPV $\approx 27,340 > 0$: accept

3. Compound Frequency

$$A = P(1 + r/n)^{nt}$$

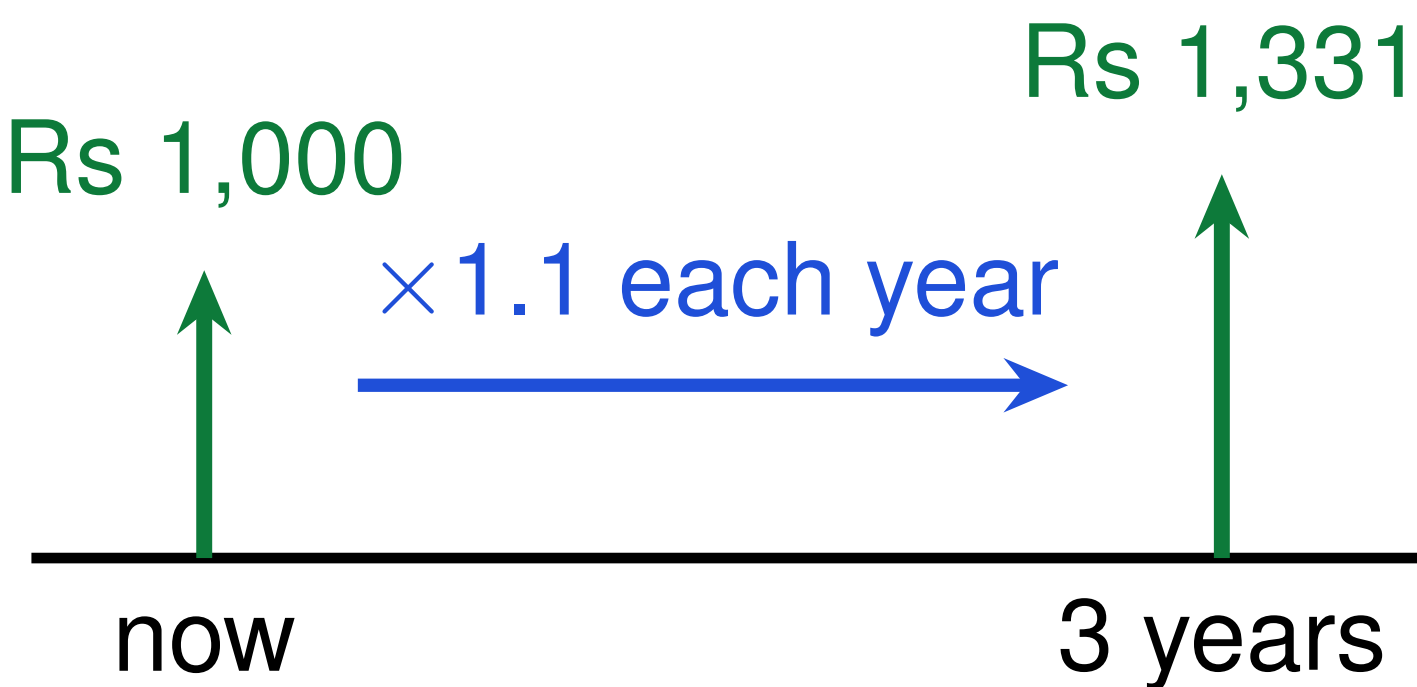
Continuous: $A = P e^{rt}$
Rs 5,00,000 at 8%, 1 year
(to nearest Rs)

How often	Rs after 1 yr
Yearly	5,40,000
Quarterly	5,41,216
Monthly	5,41,500
Continuous	5,41,644

4. Equivalence

$$FV = PV(1 + i)^n$$

At 10%: Rs 1,000 now
 $\equiv$ Rs 1,331 after 3 years



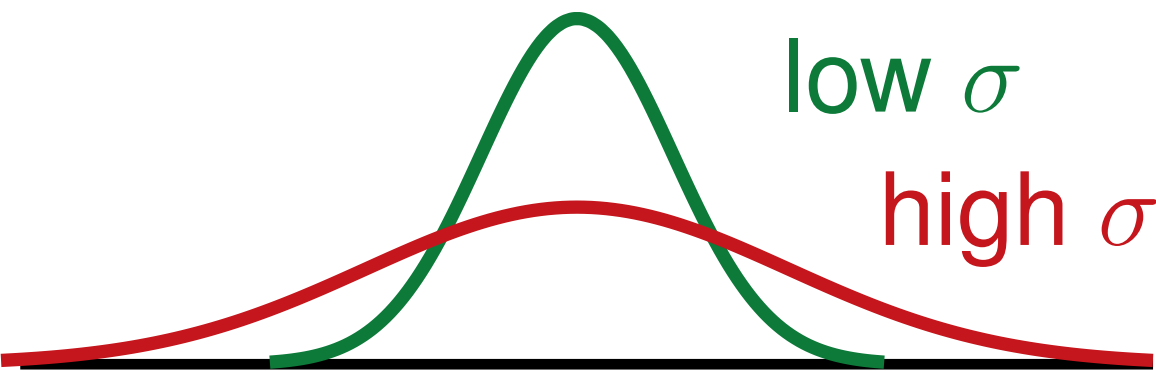
UNIT VI: RISK AND UNCERTAINTY

1. Risk: σ and CV

$$\sigma = \sqrt{\sum d^2 \div n}$$

$$\text{CV} = \sigma \div \text{mean} \times 100$$

12, 18, 20, 15, 25 (%)
 $\sigma = 4.43\%$, CV = 24.6%

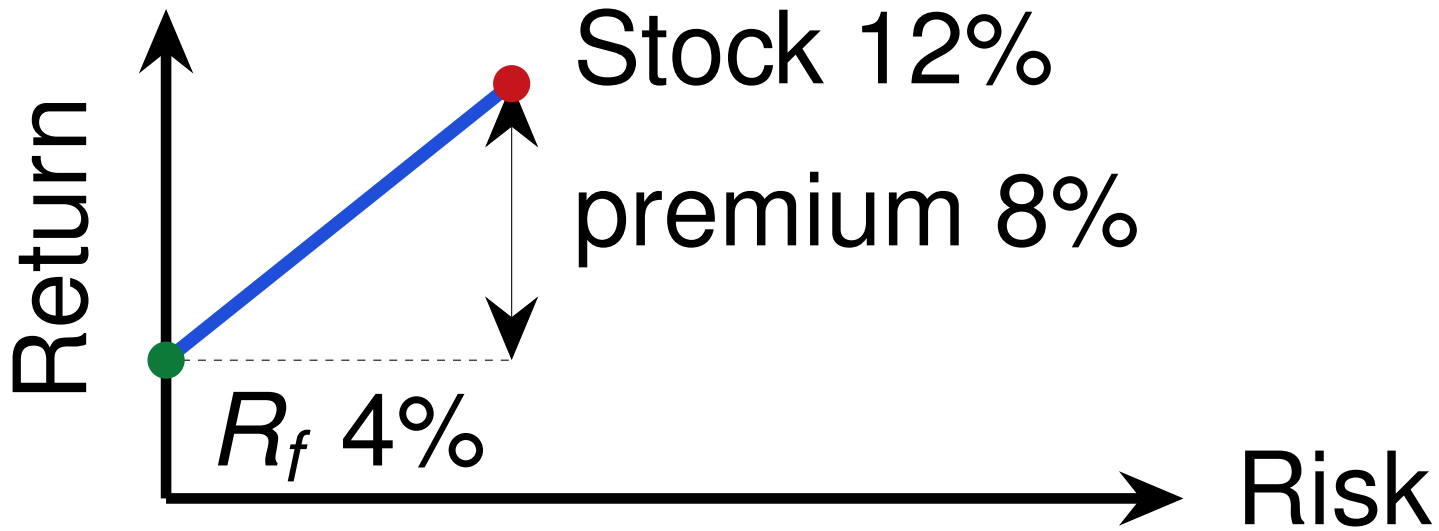


2. Risk Premium

$$\text{Premium} = E(R) - R_f$$

$$\text{Sharpe} = (R_p - R_f) \div \sigma_p$$

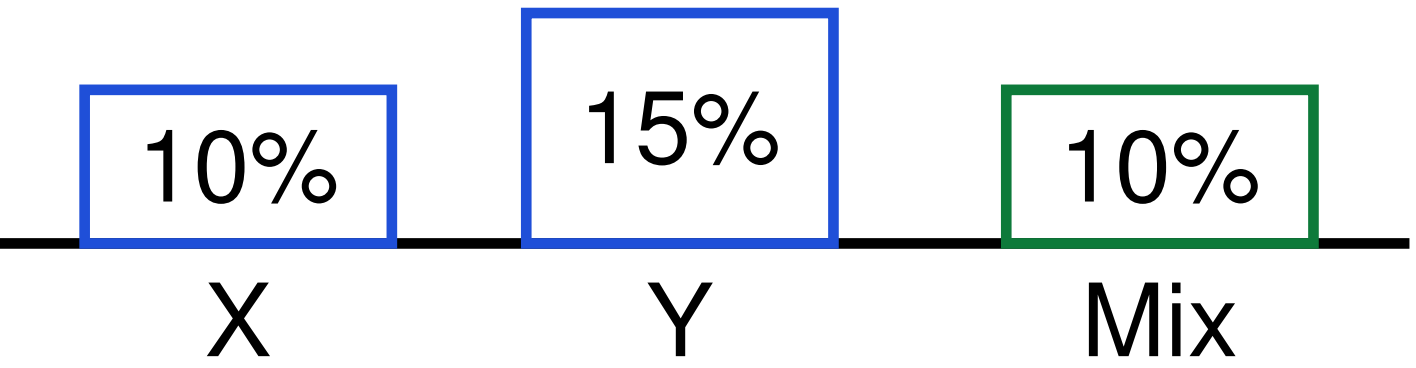
12% - 4% = 8%
Sharpe = $8 \div 10 = 0.8$ ($\sigma = 10\%$)



3. Diversification

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \sigma_1 \sigma_2 \rho$$

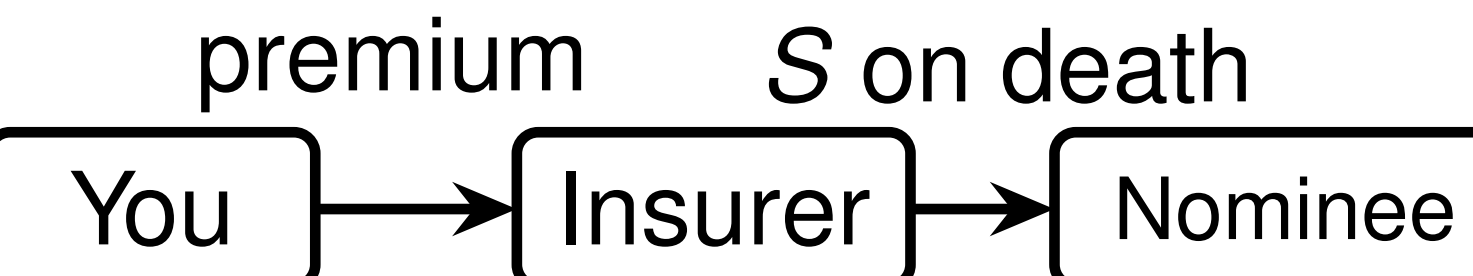
$w = 0.5, 0.5$; $\sigma = 10\%, 15\%$
 $\rho = 0.25$: $\sigma_p^2 = 0.01$



4. Life Insurance

$$\text{APV} = S \sum v^k P(T = k)$$

3-yr term, $S = 2,00,000, i = 5\%$
 $P(T=1, 2, 3) = 0.003, 0.0025, 0.002$
APV = Rs 1,370.48



5. Endowment

Term + Pure endowment



Higher risk should
earn a higher
expected return